

## Education

### Monash University

Department of Econometrics and  
Business Statistics

Doctor of Philosophy

June 2022 - present

- Thesis title: "Investigations of Flexible Jump Modelling and Systematic Risk Implied by Option Information"
- Commenced in the University of Melbourne (June 2022) and transferred to Monash with main supervisor (from Feb 2023).

### The University of New South Wales

Department of Finance

Visiting PhD

Oct, 2024 - Dec, 2024

- Hosted by Andrew Patton

### University of Sydney

Department of Economics

Graduate Certificate in Economics Analysis

Aug 2021 - Mar 2022

- Major in Economics
- WAM: 86.5

### Monash University

Monash Business School

Bachelor of Commerce (1st Class Honors)

Feb 2017 - Dec 2020

- Major in Finance and Econometrics
- Graduated with High Distinction
- GPA: 3.9/4.0

## Job Market Paper

### Learning Financial Jumps with Infinite Gaussian Mixture

Monash University & University of Melbourne

June 2022 - Present

- **Summary:** A Bayesian nonparametric method using Dirichlet process mixture to model financial price jumps size distribution and improve the predictive distribution of returns.
- [Link to the paper](#)

## Working Papers

### Downside and Upside Systematic Risk Measure Implied by Option Information

Monash University

Nov 2024 - Present

- **Summary:** An innovative method of constructing the option implied downside and upside betas within a revised CAPM framework for systematic risk measures through option implied truncated moments.

### Optimal Probabilistic Forecasts for Risk Management

Monash University

Mar 2020 - Mar 2023

- **Summary:** This paper studies how choosing tail-focused scoring rules to calibrate forecast distributions affects risk management performance in practice. It shows that tail-focused calibration improves Value-at-Risk and expected shortfall accuracy, and also leads to better hedging outcomes when forecasting the VIX for strategies using VIX futures.
- [Link to the paper](#)

## Conference Presentations

### Bayes on the Beach

University of Wollongong, Australia

February, 2026

- Accepted

### World Congress of the Econometric Society

Seoul, South Korea

August, 2025

### The European Seminar on Bayesian Econometrics (ESOB 2025)

Melbourne Business School, Australia

August, 2025

- Poster presentation

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>The 8th International Conference on Econometrics and Statistics (EcoSta 2025)</b><br>Waseda University, Japan <ul style="list-style-type: none"> <li>Invited Talk</li> </ul> | August, 2025   |
|                                                                                                                                                                                 |                |
| <b>Annual Conference of the International Association for Applied Econometrics</b><br>University of Turin, Italy                                                                | June, 2025     |
|                                                                                                                                                                                 |                |
| <b>The 19th International Symposium on Econometric Theory and Applications: SETA 2025</b><br>University of Macau, China                                                         | June, 2025     |
|                                                                                                                                                                                 | Accepted       |
| <b>Econometric Society Australian Meetings (ESAM)</b><br>Monash University, Australia                                                                                           | December, 2024 |
|                                                                                                                                                                                 |                |
| <b>Time Series and Forecasting Symposium</b><br>University of Sydney, Australia <ul style="list-style-type: none"> <li>Best Poster Runner-up</li> </ul>                         | November, 2024 |
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## Teaching Experience

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Monash Business School</b><br>Teaching Associate <ul style="list-style-type: none"> <li>Units taught (UG/PG): <i>High Dimensional Data Analysis</i> (Head Tutor); <i>Financial Econometrics</i>; <i>Business Forecasting</i>; <i>Introductory Econometrics</i>; <i>Business Statistics</i>; <i>Data Analysis in Business</i>.</li> <li>Designed and delivered tutorials across undergraduate and postgraduate econometrics and statistics units, guiding students in implementing econometric and forecasting models in R and interpreting results.</li> <li>Coordinated tutorial teams and assessment processes for <i>High Dimensional Data Analysis</i>.</li> <li>Recipient of Excellent Teaching Team Awards (<i>Data Analysis in Business</i>, 2024; <i>Business Forecasting</i>, 2025) and consistently achieved top-tier student evaluation outcomes.</li> </ul> | Melbourne, Australia |
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## Work Experience

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| <b>Monash Business School</b><br>Research Assistant <ul style="list-style-type: none"> <li>Conducted empirical analysis on high-frequency bond market data to quantify market behaviour and assess the impact of regulatory and policy-related changes;</li> <li>Built clean, reproducible workflows for data preparation and diagnostics, improving reliability of outputs across large datasets.</li> <li>Delivered timely statistical modelling and synthesised quantitative findings into clear summaries to support decision-making.</li> </ul>                                                                                               | Melbourne, Australia |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Aug 2023 - Oct 2024  |
| <b>Melbourne Business School</b><br>Research Assistant <ul style="list-style-type: none"> <li>Automated large-scale high-frequency equity data retrieval and processing pipelines using Linux language, Python and R for millisecond return of 500 stocks, improving efficiency and reproducibility of dataset construction;</li> <li>Cleaned and validated high-frequency time-series data, implemented quality checks, and produced analysis-ready datasets to support modelling and forecasting tasks;</li> <li>Collaborated with academic stakeholders to translate project requirements into structured datasets and deliverables.</li> </ul> | Melbourne, Australia |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Aug 2022 - Jan 2023  |
| <b>Liaowang Think Tank - Xinhua News Agency</b><br>Research Analyst <ul style="list-style-type: none"> <li>Collaborated with small teams and contributed to policy-focused research projects by forming hypotheses, conducting macroeconomic analysis, and synthesising results into practical recommendations;</li> <li>Produced written reports and presentation-ready materials for non-technical audiences, strengthening stakeholder communication and clarity of insights.</li> </ul>                                                                                                                                                        | Beijing, China       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | July 2021 - Oct 2021 |

## Research Grants

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|-------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Prestigious International Conference Award</b><br>Monash Business School, 2025                     | 2,300 AUD                        |
|                                                                                                       |                                  |
| <b>CFM/SoFiE Scholarship</b><br>Capital Fund Management, The Society for Financial Econometrics, 2025 | Conference Accommodation Support |
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## Honors & Awards

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|------------|-------------------------------------------------------------------------------------------------------|
| 2024, 2025 | <b>Excellent Teaching Team</b> , Monash Business School                                               |
| 2024       | <b>Best Poster Runner Up</b> , Time Series and Forecasting Symposium, University of Sydney            |
| 2023       | <b>Co-funded Monash Graduate Scholarship</b> , Monash University                                      |
| 2023       | <b>Monash Business School Top-Up Scholarship</b> , Monash Business School                             |
| 2022       | <b>Faculty of Business and Economics Graduate Research Scholarship</b> , University of Melbourne      |
| 2021       | <b>2020 Dean’s Honor (First Class)</b> , Monash Business School                                       |
| 2020       | <b>Econometrics Memorial Scholarship</b> , Department of Econometrics and Business Statistics, Monash |
| 2020       | <b>Meritorious Winner</b> , 2020 Interdisciplinary contest in Modelling                               |

## Skills

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**Programming** R, MATLAB, VBA, Python, Linux, Git, Excel

## Referees

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|                           |                               |
|---------------------------|-------------------------------|
| <b>Ole Maneesoonthorn</b> | Ole.Maneesoonthorn@monash.edu |
| <b>Yong Song</b>          | yong.song@unimelb.edu.au      |
| <b>Wei Wei</b>            | Wei.Wei2@monash.edu           |